Aishwarya Hugar

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SUMMARY

GenAI, AI/ML & Data Engineer with 4 years of experience designing and deploying RAG, NLP and agentic AI solutions, along with MLOps workflows and scalable data pipelines. Skilled in Python, SQL, PySpark, Snowflake, AWS, Airflow, LangChain, Hugging Face, MLflow and Docker.

EXPERIENCE

AI/ML Engineer — Freelance / Contract Nov 2025 – Present
Client GenAI builds across consulting, sales and GTM workflows. Pune

- Built an **adaptive, LLM-driven requirement-discovery platform** (Python, RAG, AWS) used across **8 client engagements**, raising questionnaire completion from **~25% to 80%** and cutting requirement-gathering time by **~60%**.
- Built an **agentic sales/CRM assistant** (lead tracking, follow-up nudges, automated CRM entry) from POC to rollout; adopted weekly by **9 of 12 reps**, saving an estimated **~5 hours per rep per week**.
- Designed and deployed **7+ AI agents on n8n** with **MCP** integrations to automate GTM, research and operational workflows.
- Built an **LLM evaluation system** for continuous testing and optimisation of agent outputs, improving agent output efficiency by **50%**.
- Built client-specific product prototypes that increased **sales-demo conversion by 25%**.
- Ran **build-vs-buy evaluations** to choose the right AI approach for each client requirement and expected business value.

Assistant Manager, ML/AI — SarvaGram Feb 2023 – Nov 2025
End-to-end GenAI/NLP product delivery in BFSI. Pune

- Owned end-to-end delivery of **GenAI/NLP products** across BFSI and marketing, from requirements through production deployment.
- Built a **RAG knowledge assistant for ~3,000 employees across 6 teams**, achieving **93% answer accuracy, 91% groundedness and <4% hallucination**.
- Drove **76% first-session activation and 72% 30-day retention**, while reducing manual HR/sales policy lookups by **~40%**.
- Built a Voice-of-Customer NLP pipeline processing **~12,000 feedback items** at **87% classification accuracy**; insights informed **3 product improvements**.
- Led development of an ML-based EWS to flag high-risk borrowers, improving **early-risk detection by 20%** and enabling proactive collections.

Career Break Nov 2018 – Jan 2023
Pune

Assistant Manager — ICICI Bank Jul 2016 – Oct 2018
Customer and operations analytics. Pune

- Analysed customer and operations data to deliver insights for **zonal review meetings**, helping leadership prioritise high-impact operational improvements.

Internship — Tata Consultancy Services Dec 2015 – May 2016
Data Analysis Pune

EDUCATION

Bachelor of Science, Information Technology 2013 – 2016
BVDU, Pune

PROJECTS

Project 1: GenAI Knowledge Assistant (RAG / Agentic AI)

- Designed a secure GenAI assistant for operations and collections teams to query SOPs, loan rules, customer summaries and reporting data in natural language.
- Built **RAG pipelines** over SOPs, FAQs, escalation policies, historical resolutions and data dictionaries using embeddings and vector search for grounded answers.
- Implemented **LangGraph routing** across document retrieval, NL-to-SQL, business-rule lookup, answer generation and fallback human escalation.
- Generated policy-grounded answers, collection follow-up recommendations, customer summary notes and downloadable PDF action reports.
- Developed **FastAPI** services with prompt templates, role-based access control, response validation, audit logging and monitoring hooks.

Tools & Technologies: Python, FastAPI, LangChain, LangGraph, RAG, Hugging Face, Vector Database, Snowflake, PostgreSQL, AWS S3, Airflow, Docker, MLflow

Project 2: Multilingual Call Intelligence & Sentiment Pipeline (AI/NLP + MLOps)

- Designed a speech-to-sentiment pipeline for **5 Indian languages** using Whisper-large-v3, IndicBERT/MuRIL, Hugging Face and **LoRA fine-tuning**.
- Built the workflow covering audio ingestion, preprocessing, ASR, language detection, sentiment/intent prediction, risk classification, human review, retraining and serving.
- Used **RAG-driven policy retrieval** to recommend next actions on collection calls from SOPs, escalation rules and predicted repayment intent.
- Implemented an **MLOps loop** with Airflow, MLflow, DVC, Label Studio and SageMaker for experiment tracking, dataset versioning, model promotion and monitoring.

Tools & Technologies: Python, Whisper-large-v3, MuRIL, Hugging Face, LoRA, PyTorch, Airflow, MLflow, DVC, Label Studio, AWS S3, SageMaker

Project 3: Product Review Intelligence & Insight Generator (GenAI / NLP)

- Built a review intelligence workflow using **RoBERTa**, embeddings and LLM summarisation to convert customer feedback into structured product insights.
- Combined aspect-based sentiment, keyphrase extraction and prompt-based generation to produce feature-level summaries, pain-point briefs and improvement suggestions.

Tools & Technologies: Python, RoBERTa, Hugging Face, Scikit-learn, Pandas, Embeddings, Keyphrase Extraction, LLM Summarisation

Project 4: Customer Propensity & Campaign Recommendation Generator (ML + GenAI)

- Developed **Random Forest and XGBoost** models for propensity and cross-sell scoring on transactional and demographic features; monitored uplift.
- Generated ranked customer segments with explainability notes, next-best-offer summaries and campaign-ready recommendation files.

Tools & Technologies: Python, SQL, Scikit-learn, XGBoost, Pandas, Snowflake

Project 5: Snowflake Migration & Real-Time Ingestion

- Designed S3-based landing and **Snowpipe** ingestion for near real-time loads into Snowflake.
- Implemented **PySpark** transformations in AWS Glue with SQS event-driven monitoring; optimised clustering and query performance with Snowflake admins.

Tools & Technologies: Snowflake, Snowpipe, AWS S3, AWS Glue, PySpark, Amazon SQS, SQL

SKILLS

- **GenAI & LLM:** RAG, LangChain, LangGraph, Agentic & Multi-agent Workflows, MCP, n8n, Hugging Face, LoRA Fine-tuning, LLM Evaluation, Guardrails, Prompt Engineering
- **ML & NLP:** Scikit-learn, XGBoost, PyTorch, RoBERTa, IndicBERT/MuRIL, Whisper, Predictive Modelling
- **Data:** Python, SQL, Pandas, NumPy, PySpark, Snowflake, PostgreSQL, Databricks, Power BI
- **Cloud & MLOps:** AWS (S3, Glue, SageMaker, SQS, Bedrock AgentCore), Azure AI Foundry, Docker, FastAPI, Airflow, MLflow, DVC
- **Tools:** Git, Postman, Jira, Label Studio